

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} 0 & G_y & 0 \end{bmatrix}$$

$$G_y = \mathbb{G}_y^{(1)}$$

* Rank 2 tensor (s).

$$\begin{bmatrix} 0 & G_{xy} & 0 \\ G_{xy} & 0 & G_{yz} \\ 0 & G_{yz} & 0 \end{bmatrix}$$

$$G_{xy} = \mathbb{G}_{xy}^{(1)}$$

$$G_{yz} = \mathbb{G}_{yz}^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & G_{xy} & 0 \\ -G_{xy} & 0 & G_{yz} \\ 0 & -G_{yz} & 0 \end{bmatrix}$$

$$G_{xy} = \mathbb{Q}_z^{(1)}$$

$$G_{yz} = \mathbb{Q}_x^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & G_{xxy} & 0 \\ 0 & G_{yyy} & 0 \\ 0 & G_{zzz} & 0 \\ G_{yzx} & 0 & G_{zzy} \\ 0 & G_{yzx} & 0 \\ G_{xxy} & 0 & G_{yzx} \end{bmatrix}$$

$$G_{xxy} = \mathbb{G}_y^{(1)} - \mathbb{G}_y^{\alpha(1)} - \mathbb{G}_y^{\beta(1)}$$

$$G_{yyy} = 3\mathbb{G}_y^{(1)} + 2\mathbb{G}_y^{\alpha(1)}$$

$$G_{zzz} = \mathbb{G}_y^{(1)} - \mathbb{G}_y^{\alpha(1)} + \mathbb{G}_y^{\beta(1)}$$

$$G_{yzx} = \mathbb{G}_{xyz}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & G_{xxy} & 0 \\ 0 & 0 & 0 \\ 0 & G_{zzz} & 0 \\ G_{yzx} & 0 & -\frac{G_{zzz}}{2} \\ 0 & G_{zxy} & 0 \\ -\frac{G_{xxy}}{2} & 0 & -G_{yzx} - G_{zxy} \end{bmatrix}$$

$$G_{xxy} = 2\mathbb{G}_y^{(2)} - 2\mathbb{Q}_{xz}^{(1)}$$

$$G_{zzz} = 2\mathbb{G}_y^{(2)} + 2\mathbb{Q}_{xz}^{(1)}$$

$$G_{yzx} = 3\mathbb{Q}_u^{(1)} + \mathbb{Q}_v^{(1)}$$

$$\frac{G_{zzz}}{2} = \mathbb{G}_y^{(2)} + \mathbb{Q}_{xz}^{(1)}$$

$$G_{zxy} = -3\mathbb{Q}_u^{(1)} + \mathbb{Q}_v^{(1)}$$

$$\frac{G_{xxy}}{2} = \mathbb{G}_y^{(2)} - \mathbb{Q}_{xz}^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} G_{yzx} & 0 & G_{yzz} \\ 0 & G_{zxy} & 0 \\ G_{xyx} & 0 & G_{xyz} \end{bmatrix}$$

$$G_{yzx} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_u^{(2)} + \mathbb{Q}_v^{(2)}$$

$$G_{yyz} = -\mathbb{G}_y^{(3)} + \mathbb{Q}_{xz}^{(2)}$$

$$G_{zxy} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_u^{(2)} - \mathbb{Q}_v^{(2)}$$

$$G_{xyx} = \mathbb{G}_y^{(3)} + \mathbb{Q}_{xz}^{(2)}$$

$$G_{xyz} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_u^{(2)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & G_{xxyz} & 0 & G_{xxxy} \\ 0 & 0 & 0 & G_{yyyz} & 0 & G_{yyxy} \\ 0 & 0 & 0 & G_{zzyz} & 0 & G_{zzxy} \\ G_{xxyz} & G_{yyyz} & G_{zzyz} & 0 & G_{yzzx} & 0 \\ 0 & 0 & 0 & G_{yzzx} & 0 & G_{zxxxy} \\ G_{xxxy} & G_{yyxy} & G_{zzxy} & 0 & G_{zxxxy} & 0 \end{bmatrix}$$

$$G_{xxyz} = 2\mathbb{G}_x^{\beta(1)} + \mathbb{G}_{yz}^{(1)}$$

$$G_{xxxy} = \mathbb{G}_{4z}^{\alpha(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} - \mathbb{G}_z^{\beta(1)}$$

$$G_{yyyz} = \mathbb{G}_{4x}^{\alpha(1)} - \mathbb{G}_x^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + 2\mathbb{G}_{yz}^{(2)}$$

$$G_{yyxy} = -\mathbb{G}_{4z}^{\alpha(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} - \mathbb{G}_z^{\beta(1)}$$

$$G_{zzyz} = -\mathbb{G}_{4x}^{\alpha(1)} - \mathbb{G}_x^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + 2\mathbb{G}_{yz}^{(2)}$$

$$G_{zzxy} = \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_z^{\beta(1)}$$

$$G_{yzzx} = \mathbb{G}_{xy}^{(2)} + 2\mathbb{G}_z^{\beta(1)}$$

$$G_{zxxxy} = 2\mathbb{G}_x^{\beta(1)} + \mathbb{G}_{yz}^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & G_{xxyz} & 0 & G_{xxxy} \\ 0 & 0 & 0 & G_{yyyz} & 0 & G_{yyxy} \\ 0 & 0 & 0 & G_{zzyz} & 0 & G_{zzxy} \\ -G_{xxyz} & -G_{yyyz} & -G_{zzyz} & 0 & G_{yzzx} & 0 \\ 0 & 0 & 0 & -G_{yzzx} & 0 & G_{zxxxy} \\ -G_{xxxy} & -G_{yyxy} & -G_{zzxy} & 0 & -G_{zxxxy} & 0 \end{bmatrix}$$

$$G_{xxyz} = \mathbb{G}_{yz}^{(3)} - 4\mathbb{Q}_x^{\beta(1)}$$

$$G_{xxxy} = \mathbb{G}_{xy}^{(3)} + 2\mathbb{Q}_z^{(1)} - 2\mathbb{Q}_z^{\alpha(1)} + 2\mathbb{Q}_z^{\beta(1)}$$

$$G_{yyyz} = \mathbb{G}_{yz}^{(3)} + 2\mathbb{Q}_x^{(1)} - 2\mathbb{Q}_x^{\alpha(1)} + 2\mathbb{Q}_x^{\beta(1)}$$

$$G_{yyxy} = \mathbb{G}_{xy}^{(3)} - 2\mathbb{Q}_z^{(1)} + 2\mathbb{Q}_z^{\alpha(1)} + 2\mathbb{Q}_z^{\beta(1)}$$

$$G_{zzyz} = \mathbb{G}_{yz}^{(3)} - 2\mathbb{Q}_x^{(1)} + 2\mathbb{Q}_x^{\alpha(1)} + 2\mathbb{Q}_x^{\beta(1)}$$

$$G_{zzxy} = \mathbb{G}_{xy}^{(3)} - 4\mathbb{Q}_z^{\beta(1)}$$

$$G_{yzzx} = -\mathbb{Q}_z^{(1)} - 4\mathbb{Q}_z^{\alpha(1)}$$

$$G_{zxxxy} = -\mathbb{Q}_x^{(1)} - 4\mathbb{Q}_x^{\alpha(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} 0 & G_{yzzx} & 0 \\ G_{yzzx} & 0 & G_{zxxxy} \\ 0 & G_{zxxxy} & 0 \end{bmatrix}$$

$$G_{yzzx} = 2\mathbb{G}_{xy}^{(4)}$$

$$G_{zxxy} = 2\mathbb{G}_{yz}^{(4)}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & G_{yzzx} & 0 \\ -G_{yzzx} & 0 & G_{zxxy} \\ 0 & -G_{zxxy} & 0 \end{bmatrix}$$

$$G_{yzzx} = -\mathbb{Q}_z^{(2)}$$

$$G_{zxxy} = -\mathbb{Q}_x^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} G_{xxyz} & 0 & G_{xxxy} \\ G_{yyyz} & 0 & G_{yyxy} \\ G_{zzyz} & 0 & G_{zzxy} \\ 0 & G_{yzzx} & 0 \\ G_{zxyz} & 0 & G_{zxxy} \\ 0 & G_{xyzx} & 0 \end{bmatrix}$$

$$G_{xxyz} = \mathbb{Q}_x^{(3)} + 2\mathbb{Q}_x^{(4)} + 2\mathbb{Q}_x^{\alpha(2)}$$

$$G_{xxxy} = -2\mathbb{G}_{xy}^{(5)} + \mathbb{Q}_z^{(3)} - \mathbb{Q}_z^{\alpha(2)} + \mathbb{Q}_z^{\beta(2)}$$

$$G_{yyyz} = -2\mathbb{G}_{yz}^{(5)} + \mathbb{Q}_x^{(3)} - \mathbb{Q}_x^{\alpha(2)} + \mathbb{Q}_x^{\beta(2)}$$

$$G_{yyxy} = 2\mathbb{G}_{xy}^{(5)} + \mathbb{Q}_z^{(3)} - \mathbb{Q}_z^{\alpha(2)} - \mathbb{Q}_z^{\beta(2)}$$

$$G_{zzyz} = 2\mathbb{G}_{yz}^{(5)} + \mathbb{Q}_x^{(3)} - \mathbb{Q}_x^{\alpha(2)} - \mathbb{Q}_x^{\beta(2)}$$

$$G_{zzxy} = \mathbb{Q}_z^{(3)} + 2\mathbb{Q}_z^{(4)} + 2\mathbb{Q}_z^{\alpha(2)}$$

$$G_{yzzx} = -\mathbb{G}_{xy}^{(5)} + \mathbb{Q}_z^{(4)} - \mathbb{Q}_z^{\alpha(2)} - \mathbb{Q}_z^{\beta(2)}$$

$$G_{zxyz} = \mathbb{G}_{xy}^{(5)} + \mathbb{Q}_z^{(4)} - \mathbb{Q}_z^{\alpha(2)} + \mathbb{Q}_z^{\beta(2)}$$

$$G_{zxxy} = -\mathbb{G}_{yz}^{(5)} + \mathbb{Q}_x^{(4)} - \mathbb{Q}_x^{\alpha(2)} - \mathbb{Q}_x^{\beta(2)}$$

$$G_{xyzx} = \mathbb{G}_{yz}^{(5)} + \mathbb{Q}_x^{(4)} - \mathbb{Q}_x^{\alpha(2)} + \mathbb{Q}_x^{\beta(2)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} G_{yzxx} & G_{yzyy} & G_{yzzz} & 0 & G_{yzzx} & 0 \\ 0 & 0 & 0 & G_{zxyz} & 0 & G_{zxxy} \\ G_{xyxx} & G_{xyyy} & G_{xyzz} & 0 & G_{xyzx} & 0 \end{bmatrix}$$

$$G_{yzxx} = \mathbb{Q}_x^{(5)} + 2\mathbb{Q}_x^{(6)} + 2\mathbb{Q}_x^{\alpha(3)}$$

$$G_{yzyy} = -2\mathbb{G}_{yz}^{(6)} + \mathbb{Q}_x^{(5)} - \mathbb{Q}_x^{\alpha(3)} + \mathbb{Q}_x^{\beta(3)}$$

$$G_{yzzz} = 2\mathbb{G}_{yz}^{(6)} + \mathbb{Q}_x^{(5)} - \mathbb{Q}_x^{\alpha(3)} - \mathbb{Q}_x^{\beta(3)}$$

$$G_{yzzx} = \mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{(6)} - \mathbb{Q}_z^{\alpha(3)} + \mathbb{Q}_z^{\beta(3)}$$

$$G_{zxyz} = -\mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{(6)} - \mathbb{Q}_z^{\alpha(3)} - \mathbb{Q}_z^{\beta(3)}$$

$$G_{zxxy} = \mathbb{G}_{yz}^{(6)} + \mathbb{Q}_x^{(6)} - \mathbb{Q}_x^{\alpha(3)} + \mathbb{Q}_x^{\beta(3)}$$

$$G_{xyxx} = -2\mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{(5)} - \mathbb{Q}_z^{\alpha(3)} + \mathbb{Q}_z^{\beta(3)}$$

$$G_{xyyy} = 2\mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{(5)} - \mathbb{Q}_z^{\alpha(3)} - \mathbb{Q}_z^{\beta(3)}$$

$$G_{xyzz} = \mathbb{Q}_z^{(5)} + 2\mathbb{Q}_z^{(6)} + 2\mathbb{Q}_z^{\alpha(3)}$$

$$G_{xyzx} = -\mathbb{G}_{yz}^{(6)} + \mathbb{Q}_x^{(6)} - \mathbb{Q}_x^{\alpha(3)} - \mathbb{Q}_x^{\beta(3)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & G_{xyz} & 0 & G_{xxy} & G_{xzy} & 0 & G_{xyx} \\ 0 & 0 & 0 & G_{yyz} & 0 & G_{yyx} & G_{yyz} & 0 & G_{yyy} \\ 0 & 0 & 0 & G_{zzz} & 0 & G_{zzx} & G_{zzz} & 0 & G_{zzx} \\ G_{yzx} & G_{zyy} & G_{yzz} & 0 & G_{yzz} & 0 & 0 & G_{yzx} & 0 \\ 0 & 0 & 0 & G_{zyz} & 0 & G_{zxy} & G_{zzy} & 0 & G_{zzy} \\ G_{yxx} & G_{xyy} & G_{xyz} & 0 & G_{yxz} & 0 & 0 & G_{yxz} & 0 \end{bmatrix}$$

[illegible]

* Rank 4 tensor (a).

$$\begin{bmatrix} G_{yzzx} & G_{yzzy} & G_{yzzz} & 0 & G_{yzxz} & 0 & 0 & G_{yzzx} & 0 \\ 0 & 0 & 0 & G_{zyyz} & 0 & G_{zyxy} & G_{zyxz} & 0 & G_{zyyx} \\ G_{xyxx} & G_{xyyy} & G_{xyzz} & 0 & G_{xyzx} & 0 & 0 & G_{xyxz} & 0 \end{bmatrix}$$

$$\begin{aligned}
G_{yzxx} &= \mathbb{Q}_x^{(5)} + 2\mathbb{Q}_x^{(6)} + 2\mathbb{Q}_x^{\alpha(3)} \\
G_{yzyy} &= -2\mathbb{G}_{yz}^{(6)} + \mathbb{Q}_x^{(5)} - \mathbb{Q}_x^{\alpha(3)} + \mathbb{Q}_x^{\beta(3)} \\
G_{yzzz} &= 2\mathbb{G}_{yz}^{(6)} + \mathbb{Q}_x^{(5)} - \mathbb{Q}_x^{\alpha(3)} - \mathbb{Q}_x^{\beta(3)} \\
G_{yzzx} &= 2\mathbb{G}_{xy}^{(4)} + \mathbb{G}_{xy}^{(6)} - \mathbb{Q}_z^{(2)} + \mathbb{Q}_z^{(6)} - \mathbb{Q}_z^{\alpha(3)} + \mathbb{Q}_z^{\beta(3)} \\
G_{yzxz} &= -2\mathbb{G}_{xy}^{(4)} + \mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{(2)} + \mathbb{Q}_z^{(6)} - \mathbb{Q}_z^{\alpha(3)} + \mathbb{Q}_z^{\beta(3)} \\
G_{zxyy} &= 2\mathbb{G}_{xy}^{(4)} - \mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{(2)} + \mathbb{Q}_z^{(6)} - \mathbb{Q}_z^{\alpha(3)} - \mathbb{Q}_z^{\beta(3)} \\
G_{zxxy} &= 2\mathbb{G}_{yz}^{(4)} + \mathbb{G}_{yz}^{(6)} - \mathbb{Q}_x^{(2)} + \mathbb{Q}_x^{(6)} - \mathbb{Q}_x^{\alpha(3)} + \mathbb{Q}_x^{\beta(3)} \\
G_{zxxz} &= -2\mathbb{G}_{xy}^{(4)} - \mathbb{G}_{xy}^{(6)} - \mathbb{Q}_z^{(2)} + \mathbb{Q}_z^{(6)} - \mathbb{Q}_z^{\alpha(3)} - \mathbb{Q}_z^{\beta(3)} \\
G_{zxyx} &= -2\mathbb{G}_{yz}^{(4)} + \mathbb{G}_{yz}^{(6)} + \mathbb{Q}_x^{(2)} + \mathbb{Q}_x^{(6)} - \mathbb{Q}_x^{\alpha(3)} + \mathbb{Q}_x^{\beta(3)} \\
G_{xyxx} &= -2\mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{(5)} - \mathbb{Q}_z^{\alpha(3)} + \mathbb{Q}_z^{\beta(3)} \\
G_{xyyy} &= 2\mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{(5)} - \mathbb{Q}_z^{\alpha(3)} - \mathbb{Q}_z^{\beta(3)} \\
G_{xyzx} &= \mathbb{Q}_z^{(5)} + 2\mathbb{Q}_z^{(6)} + 2\mathbb{Q}_z^{\alpha(3)} \\
G_{xyzx} &= 2\mathbb{G}_{yz}^{(4)} - \mathbb{G}_{yz}^{(6)} + \mathbb{Q}_x^{(2)} + \mathbb{Q}_x^{(6)} - \mathbb{Q}_x^{\alpha(3)} - \mathbb{Q}_x^{\beta(3)} \\
G_{xyxz} &= -2\mathbb{G}_{yz}^{(4)} - \mathbb{G}_{yz}^{(6)} - \mathbb{Q}_x^{(2)} + \mathbb{Q}_x^{(6)} - \mathbb{Q}_x^{\alpha(3)} - \mathbb{Q}_x^{\beta(3)}
\end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix}
0 & G_{xxxy} & 0 \\
G_{yyyy} & 0 & G_{yyyz} \\
0 & G_{zzzy} & 0 \\
0 & G_{yyzy} & 0 \\
0 & G_{zzxy} & 0 \\
G_{xxyx} & 0 & G_{xxyz} \\
G_{yzzx} & 0 & G_{yzzz} \\
0 & G_{zxxy} & 0 \\
0 & G_{xyyy} & 0 \\
-\frac{G_{yyyyz}}{2} + \frac{G_{yyzy}}{2} + G_{zxxy} & 0 & \frac{G_{xxxxy}}{2} - \frac{G_{xxyx}}{2} + G_{xyyy} - G_{yyyx} + G_{zzxy}
\end{bmatrix}$$

$$\begin{aligned}
G_{xxxy} &= \mathbb{G}_{4z}^{\alpha(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} - 2\mathbb{G}_{xy}^{(5)} - \mathbb{G}_z^{\beta(1)} \\
G_{yyyy} &= -\mathbb{G}_{4z}^{\alpha(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} - 2\mathbb{G}_{xy}^{(5)} - \mathbb{G}_z^{\beta(1)} \\
G_{yyyz} &= \mathbb{G}_{4x}^{\alpha(1)} - \mathbb{G}_x^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + 2\mathbb{G}_{yz}^{(2)} + \mathbb{G}_{yz}^{(3)} - 2\mathbb{G}_{yz}^{(5)} \\
G_{zzzy} &= -\mathbb{G}_{4x}^{\alpha(1)} - \mathbb{G}_x^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + 2\mathbb{G}_{yz}^{(2)} + \mathbb{G}_{yz}^{(3)} - 2\mathbb{G}_{yz}^{(5)} \\
G_{yyzy} &= \mathbb{G}_{4x}^{\alpha(1)} - \mathbb{G}_x^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + 2\mathbb{G}_{yz}^{(2)} + \mathbb{G}_{yz}^{(3)} + 2\mathbb{G}_{yz}^{(5)} \\
G_{zzxy} &= \mathbb{G}_{xy}^{(1)} + \mathbb{G}_{xy}^{(3)} + 2\mathbb{G}_z^{\beta(1)} \\
G_{xxyx} &= \mathbb{G}_{4z}^{\alpha(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} + 2\mathbb{G}_{xy}^{(5)} - \mathbb{G}_z^{\beta(1)} \\
G_{xxyz} &= 2\mathbb{G}_x^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + \mathbb{G}_{yz}^{(3)} \\
G_{yzzx} &= \mathbb{G}_{xy}^{(2)} - \mathbb{G}_{xy}^{(5)} + 2\mathbb{G}_z^{\beta(1)} \\
G_{yzzz} &= -\mathbb{G}_{4x}^{\alpha(1)} - \mathbb{G}_x^{\beta(1)} + \mathbb{G}_{yz}^{(1)} + 2\mathbb{G}_{yz}^{(2)} - \mathbb{G}_{yz}^{(3)} \\
G_{zxxxy} &= 2\mathbb{G}_x^{\beta(1)} + \mathbb{G}_{yz}^{(2)} - \mathbb{G}_{yz}^{(5)} \\
G_{xyyy} &= -\mathbb{G}_{4z}^{\alpha(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} - \mathbb{G}_{xy}^{(3)} - \mathbb{G}_z^{\beta(1)}
\end{aligned}$$